

Yuanfan Chen

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EDUCATION

Cornell Tech, Cornell University

Master of Engineering in Computer Science

New York, NY

Aug. 2025 – May. 2026

University of Toronto

Honours Bachelor of Science, Computer Science Specialist (GPA: 3.87/4.0)

Toronto, Canada

Sep. 2020 – May. 2024

PUBLICATIONS

- **dpBento: Benchmarking DPUs for Data Processing.** Jiasheng Hu, Chihan Cui, Anna Li, Raahil Vora, **Yuanfan Chen**, Philip A. Bernstein, Jialin Li, Qizhen Zhang. arXiv:2504.05536 (2025).

PERSONAL EXPERIENCE

Machine Learning System Researcher

Cornell University

Sep 2025 – Present

New York, NY

- Designed a hierarchical **admission control mechanism** (inspired by database kernels) to manage KV-cache memory pressure, effectively preventing system congestion under bursty mixed workloads.
- Formulated a **Queueing Theory** model to characterize latency behaviors of compute-bound (prefill) and memory-bound (decode) phases, providing mathematical guarantees for multi-SLO constraints.
- Implemented a control-theoretic batch-size optimizer that dynamically adapts to real-time load, achieving **robust system stability** and a **~20% throughput gain** while strictly adhering to latency SLOs.

Kernel Development Intern

Tencent: **TCHouse-X Distributed Database**

Jul 2025 – Aug 2025

Beijing, CN

- Designed and implemented an **admission control mechanism** to regulate request concurrency, ensuring system stability under bursty loads (which later inspired my Multi-SLO research).
- Contributed to the open-source **ClickHouse-X** project; established rigorous **CI/CD pipelines** and code review standards to build robust, maintainable, and reproducible distributed systems.
- Optimized node recovery via fault-tolerant **KV delta-apply**, reducing latency from **10s to 3s** and ensuring data consistency in high-churn clusters.

Machine Learning Engineer

Tencent: **Distributed Training and Inference**

May 2025 – Sep 2025

Beijing, CN

- Mastered parallelization strategies including **Tensor Parallelism (TP)** and **Data Parallelism (DP)**, and optimized inference engines using **PagedAttention** and **Continuous Batching**.
- Orchestrated large-scale GPU tasks on **multi-node, multi-card clusters** using **Kubernetes, Ray, and DeepSpeed**, solving complex communication overheads in distributed training.
- Reduced pipeline bubbles by implementing **Chunk Prefill** and batched decode strategies, significantly improving throughput for long-context inference workloads.

Database Researcher

Far Data Lab

Jan 2024 – Feb 2025

Toronto, CA

- Co-designed **dpBento** to benchmark compute/memory/network/storage of heterogeneous DPUs for data processing offload.
- Studied **RocksDB** internals; explored DPU offload for decompression/compaction to evaluate compute-storage disaggregation.

PROJECTS

Full-Stack Distributed Deep Learning Training System

PyTorch, Cuda, Distributed Training

2025.09 – Now

- Implemented tensor storage, autograd graph, broadcasting, view semantics.
- Integrated ring-allreduce and tree-allreduce with topology-aware selection.
- Wrote CUDA GEMM (shared-memory tiling, coalesced access, occupancy tuning); optimized *im2col* → GEMM convolution.

TECHNICAL SKILLS

Programming: Python, C++, CUDA, Java **AI Infra:** vLLM, SGLang, KV-Cache, PD separation

Distributed: CAP, ClickHouse, HDFS **Databases:** RocksDB, MySQL, Redis

Networking: epoll, multiplexing, async I/O, KCP **Systems:** Linux internals, CPU/GPU arch, Nsys/Nsight